Calculating Excitons from CRYSTAL

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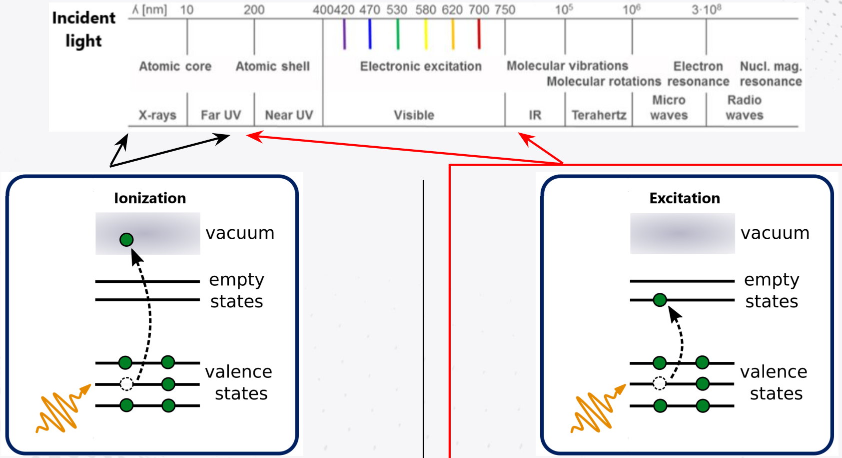
Optical processes in quantum systems



OUR FOCUS

- Measured by: photoemission spectroscopy
- Total charge **altered** ($N_{\mathrm{el}} \to N_{\mathrm{el}} - 1$)

- Measured by: absorption spectroscopy
- Total charge **conserved** ($N_{\mathrm{el}} \to N_{\mathrm{el}}$)

The effect of electronic correlations in absorption

Is the single-particle description able to accurately describe optical response?

Occasionally, yes – but overall, NO

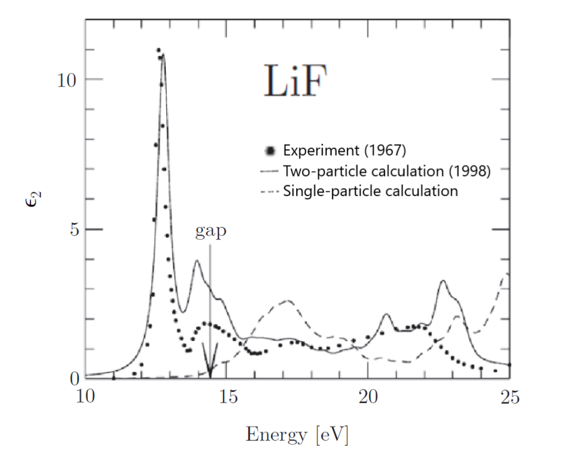


Fig.: Adapted from Martin, R. et al. *Interacting Electrons*, 2016

Correlation functions

Many-electron effects are introduced by solving the **correlated motion of two particles** $\Rightarrow$ more precise than solving for a single particle in an effective environment

Parallelism between:

- One-particle Green's function

$$G(a,b) \equiv -i\langle T[\hat{\psi}(a)\hat{\psi}^{\dagger}(b)]\rangle$$

1-electron coords.
$a \equiv (\boldsymbol{r}_a, \sigma_a, t_a)$

- Two-particle Green's function

$$G_2(a,b,a',b') \equiv -\langle T[\hat{\psi}(a)\hat{\psi}(b)\hat{\psi}^{\dagger}(b')\hat{\psi}^{\dagger}(a')]\rangle$$

	$G(a,b)$	$G_2(a,b,a',b')$
Probability amplitude for the evolution of the many-electron system going through: (depending on relative times)	$N_{\mathrm{el}} \to N_{\mathrm{el}}+1$	$N_{\mathrm{el}} \to N_{\mathrm{el}}+2$
		$N_{\mathrm{el}} \to N_{\mathrm{el}}-2$
	$N_{\mathrm{el}} \to N_{\mathrm{el}}-1$	$N_{\mathrm{el}} \to N_{\mathrm{el}}$

electron-hole ("**exciton**") sector

Many-body perturbation theory for 2 particles

Subtract uncorrelated terms to get true mathematical correlation:

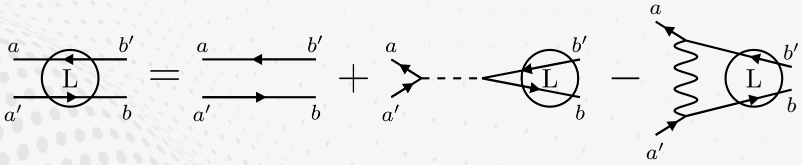
$$L(a,b,a',b') \equiv -G_2(a,b,a',b') + G(a,a')G(b,b')$$

The Dyson equation for L is the **Bethe-Salpeter equation (BSE)**:

$$L(a,b,a',b') = L_0(a,b,a',b') + L_0(a,\bar{c}',a',\bar{c})\Xi(\bar{c},\bar{d},\bar{c}',\bar{d}')L(\bar{d}',b,\bar{d},b')$$

Interaction kernel in the GW approximation (**GW-BSE**):

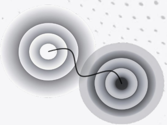
$$\Xi^{\mathrm{GW}}(a,b,a',b') = -i\delta(a,a')\delta(b'^{+},b)V(a^{+},b) + i\delta(a,b')\delta(a',b)W(a^{+},b)$$



Electron-hole Hamiltonian in solids

Exciton wavefunction (TDA):

$$\Psi_{\chi\boldsymbol{Q}}(\boldsymbol{x}_e, \boldsymbol{x}_h) = \sum_{v,c,\boldsymbol{k}} A_{\chi}^{c\boldsymbol{k}+\boldsymbol{Q},v\boldsymbol{k}} \underbrace{\psi_{c\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{x}_e)\psi^*_{v\boldsymbol{k}}(\boldsymbol{x}_h)}_{\text{exciton basis}}$$



v: valence, c: conduction

Typically $N_v, N_c \sim 10^0 - 10^1$, $N_{\boldsymbol{k}} \sim 10^3 - 10^5$ $\rightarrow$ $\ll$ energy range than in molecules

GW-BSE Hamiltonian (TD approx., $\boldsymbol{Q} = 0$, spinless)

$$H^{\text{e-h}}_{c\boldsymbol{k},v'\boldsymbol{k}';v\boldsymbol{k},c'\boldsymbol{k}'} = (\varepsilon_{c\boldsymbol{k}} - \varepsilon_{v\boldsymbol{k}})\delta_{c,c'}\delta_{v,v'}\delta_{\boldsymbol{k},\boldsymbol{k}'} + X^{v\boldsymbol{k},v'\boldsymbol{k}'}_{c\boldsymbol{k},c'\boldsymbol{k}'} - D^{c'\boldsymbol{k}',v'\boldsymbol{k}'}_{c\boldsymbol{k},v\boldsymbol{k}}$$

Exchange term:

$$X^{v\boldsymbol{k},v'\boldsymbol{k}'}_{c\boldsymbol{k},c'\boldsymbol{k}'} = \int \psi^*_{c\boldsymbol{k}}(\boldsymbol{r})\psi_{v\boldsymbol{k}}(\boldsymbol{r}) \underbrace{V(\boldsymbol{r}-\boldsymbol{r}')}_{\text{bare Coulomb}} \psi_{c'\boldsymbol{k}'}(\boldsymbol{r}')\psi^*_{v'\boldsymbol{k}'}(\boldsymbol{r}')d\boldsymbol{r}d\boldsymbol{r}' \quad = \quad \text{Hartree in TDDFT}$$

Direct term:

$$D^{c'\boldsymbol{k}',v'\boldsymbol{k}'}_{c\boldsymbol{k},v\boldsymbol{k}} = \int \psi^*_{c\boldsymbol{k}}(\boldsymbol{r})\psi_{c'\boldsymbol{k}'}(\boldsymbol{r}) \overbrace{W(\boldsymbol{r},\boldsymbol{r}')}^{\text{screened Coulomb}} \psi_{v\boldsymbol{k}}(\boldsymbol{r}')\psi^*_{v'\boldsymbol{k}'}(\boldsymbol{r}')d\boldsymbol{r}d\boldsymbol{r}' \quad \sim \quad \text{Fock exch. in hybrid TDDFT}$$

Complexity of solving the GW-BSE in a solid

$N_{\text{exc}} \equiv N_{\boldsymbol{k}} N_v N_c$ (exciton basis size) $\rightarrow$

- $H^{\text{e-h}}$ diagonalization $\sim \mathcal{O}(N_{\text{exc}}^3)$
  - Does not (necessarily) grow with system's size 😛
  - Can be reduced with Lanczos 🙂
  - Sensible to the DOS near the band edge ($\Uparrow$DOS $\Rightarrow$ $\Uparrow$exciton basis) 😐
- $H^{\text{e-h}}$ initialization (in a localized basis set with N_μ AOs)
  - Requires a lot of memory (4-center Coulomb integrals) 🥵
  - Exchange term X: $\sim \mathcal{O}(N_\mu^4 N_{\text{exc}}^2) + \ldots$ 🥵
  - Direct term D: $\sim \mathcal{O}(\underbrace{N_\mu^6}_{\text{Dyson eq for } W} N_{\text{exc}}^2) + \ldots$ 💀

The standard formulation is extremely expensive due to the screening (no analogy in TDDFT) $\Rightarrow$ only minimal bases would be suitable

Exploiting the local basis

Proceeding from CRYSTAL

1. Construct a Gaussian basis set $\{\varphi_\mu\}$ and solve for the electronic density (DFT) or density matrix (hybrid DFT) $\rightarrow$ `Pcrystal`

2. Extract the Hamiltonian ("Fock") matrices $H_{\mu,\mu'}(\boldsymbol{R}) = \left\langle \varphi_\mu^{\boldsymbol{0}} \middle| \hat{H} \middle| \varphi_{\mu'}^{\boldsymbol{R}} \right\rangle$ and overlap matrices $S_{\mu,\mu'}(\boldsymbol{R}) = \left\langle \varphi_\mu^{\boldsymbol{0}} \middle| \varphi_{\mu'}^{\boldsymbol{R}} \right\rangle$ $\rightarrow$ `properties`

.d3 file:	`BASISSET`
	`2`
	`60 N`
	`64 N`
	`END`

3. Transform to reciprocal space $H(\boldsymbol{k}) = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\boldsymbol{R}} H(\boldsymbol{R})$, $S(\boldsymbol{k}) = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\boldsymbol{R}} S(\boldsymbol{R}) \rightarrow$ single-particle spectrum via the Roothaan equations $H(\boldsymbol{k}) c_{\boldsymbol{k}} = S(\boldsymbol{k}) c_{\boldsymbol{k}} \varepsilon_{\boldsymbol{k}}$

4. Compute other matrix elements (i.e. Gaussian integrals) required by the theory and proceed

$\vdots$

Landscape of **periodic** BSE implementations

Plane-waves basis

Yambo | BerkeleyGW

exciting | WEST! | VASP | Abinit

GPAW! . . .

Localized basis

In-house Gaussian codes:

- Rohlfing, Krüger, Pollmann ($\sim$ 1995)
- C. Patterson ($\sim$ 2005)

FHI-aims
The ab initio materials simulation package

Numerical orbs. (2024)

} Only for 3D

- Present work (2024), interfaced with

CRYSTAL23

} Valid for 2D & 3D

Why use a localized basis?

Advantages of localized basis functions (vs plane-waves):

- Reduced size & real-valued $\Rightarrow$ Less operations + lower memory requirements
- Tunable (better $\not\Rightarrow$ bigger)
- Exploit sparsity in real space (Fock exchange, polarizability ...)
- No need to replicate the lattice (2D, 1D, 0D)
- Velocity and dipole matrix elements are straightforward

Disadvantages:

- Harder to approach basis completeness
- Less availability of methods / bibliography for periodic systems
- More complex control flow

For Gaussians in particular (vs other types of local bases), most integrals are analytical (i.e. much cheaper) but typically a higher number of functions is required

Resolution of the identity (RI) approx.

$$\begin{cases} \text{Crystalline orbitals: } \varphi_{\mu\boldsymbol{k}}(\boldsymbol{r}) = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\boldsymbol{R}} \varphi_{\mu}(\boldsymbol{r}-\boldsymbol{R}) \\ \textbf{Auxiliary basis of AOs: } \phi_P, \text{ with } \dfrac{\#\{\phi_P\}}{\#\{\varphi_\mu\}} \sim 2-4 \end{cases} \Longrightarrow$$

RI in reciprocal space

$$\varphi^*_{\mu\boldsymbol{k}}(\boldsymbol{r})\varphi_{\mu'\boldsymbol{k}'}(\boldsymbol{r}) \approx \sum_P B_P^{\mu\boldsymbol{k},\mu'\boldsymbol{k}'} \phi_{P\boldsymbol{k}'-\boldsymbol{k}}(\boldsymbol{r})$$

Required matrix elements:

- 2-center metric matrices

$$M_{P,P'\boldsymbol{k}}[m] = \frac{1}{N}\int \phi_{P-\boldsymbol{k}}(\boldsymbol{r}) m(\boldsymbol{r}-\boldsymbol{r}') \phi_{P'\boldsymbol{k}}(\boldsymbol{r}') d\boldsymbol{r} d\boldsymbol{r}'$$

- 3-center metric matrices

$$L_P^{\mu\boldsymbol{k},\mu'\boldsymbol{k}'}[m] = \frac{1}{N}\int \phi_{P\boldsymbol{k}-\boldsymbol{k}'}(\boldsymbol{r}) m(\boldsymbol{r}-\boldsymbol{r}') \varphi^*_{\mu\boldsymbol{k}}(\boldsymbol{r}')\varphi_{\mu'\boldsymbol{k}'}(\boldsymbol{r}') d\boldsymbol{r} d\boldsymbol{r}'$$

- 2-center Coulomb matrices $J_{\boldsymbol{k}} = M_{\boldsymbol{k}}[m(\boldsymbol{r}) = |\boldsymbol{r}|^{-1}]$

Complexity of building the GW-BSE + RI in a solid

$$\left.\begin{array}{l} N_{\mathrm{exc}} \equiv N_{\boldsymbol{k}} N_v N_c \text{ (exciton basis size)} \\ N_{\mathrm{pol}} \equiv N_{\tilde{\boldsymbol{k}}} N_{\tilde{v}} N_{\tilde{c}} \text{ (polarizability basis size)} \end{array}\right\} \rightarrow N_{\tilde{\boldsymbol{k}}} \ll N_{\boldsymbol{k}}, \quad N_{\tilde{v}}, N_{\tilde{c}} \gg N_v, N_c$$

1. Exchange term X:

$$\sim \overbrace{\mathcal{O}(N_P N_\mu^2 N_{\mathrm{exc}})}^{\sim N^3 N_{\boldsymbol{k}}} + \overbrace{\mathcal{O}(N_P N_{\mathrm{exc}}^2)}^{\sim N N_{\boldsymbol{k}}^2}$$

😵‍💫 → 😐

$+$ Cost FT of 3-center integrals to $\boldsymbol{k}$: $\propto N_P N_\mu^2 N_{\boldsymbol{k}}$ $\leftarrow$ dominates over $N^3 N_{\boldsymbol{k}}$

2. Direct term D:

$$\sim \overbrace{\mathcal{O}(N_P^2 N_{\boldsymbol{k}} N_{\mathrm{pol}})}^{\text{polarizability, } \sim N^4 N_{\boldsymbol{k}}} + \overbrace{\mathcal{O}(N_P N_\mu^2 N_{\boldsymbol{k}}^2 [N_c + N_v])}^{\sim N^3 N_{\boldsymbol{k}}^2}$$

💀 → 😵‍💫

$+$ Cost FT of 3-center integrals to $\boldsymbol{k}, \boldsymbol{k}'$: $\propto N_P N_\mu^2 N_{\boldsymbol{k}}^2$ $\leftarrow$ dominates over $N^3 N_{\boldsymbol{k}}^2$

Problems with the Coulomb lattice series

Convergence regimes (I)

2-center Coulomb matrix elements

$$J_{P,P'\boldsymbol{k}} = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\boldsymbol{R}} \int \phi_P^{\boldsymbol{0}}(\boldsymbol{r}) V(\boldsymbol{r}-\boldsymbol{r}') \phi_{P'}^{\boldsymbol{R}}(\boldsymbol{r}') d\boldsymbol{r} d\boldsymbol{r}'$$

$$\Downarrow$$

Interaction energy between: $\phi_P^{\boldsymbol{0}}$ (local) and $\rho^{\text{latt}} \equiv \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\boldsymbol{R}} \phi_{P'}^{\boldsymbol{R}}$ (extended)

k_{min}, l_{min}: lowest non-vanishing multipole moments of $\phi_P^{\boldsymbol{0}}$, ρ^{latt}

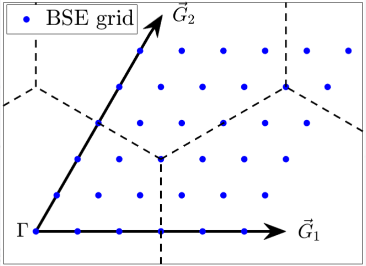
- $\sum_{\boldsymbol{R}}$ divergent if

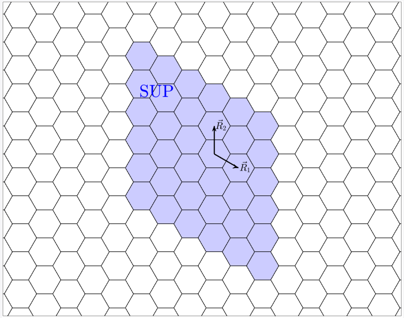
$$k_{\text{min}} + l_{\text{min}} = 0$$

Convergence regimes (II)

For $\boldsymbol{k} \neq \mathbf{0}$, ρ^{latt} is neutral and periodic in any **commensurate supercell** SUP (w.r.t. the BSE grid) $\Rightarrow$

$$k_{\text{min}} + l_{\text{min}} \geq 1$$





$$\sum_{\boldsymbol{R}}^{\text{SUP}} e^{i\boldsymbol{k}\boldsymbol{R}} = 0 \quad , \quad \forall \boldsymbol{k} \in \text{BSE grid}$$

Can the series be converged by brute force?

$$J_{P,P'\boldsymbol{k}} = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\boldsymbol{R}} \int \phi_P^{\boldsymbol{0}}(\boldsymbol{r}) V(\boldsymbol{r}-\boldsymbol{r}') \phi_{P'}^{\boldsymbol{R}}(\boldsymbol{r}') d\boldsymbol{r} d\boldsymbol{r}'$$

$\sum_{\boldsymbol{R}}$ converged $\Rightarrow J_{\boldsymbol{k}}$ is positive definite, but ...

$y \sim ax^{-1/2} + b \Rightarrow$ not even positive by extrapolation!

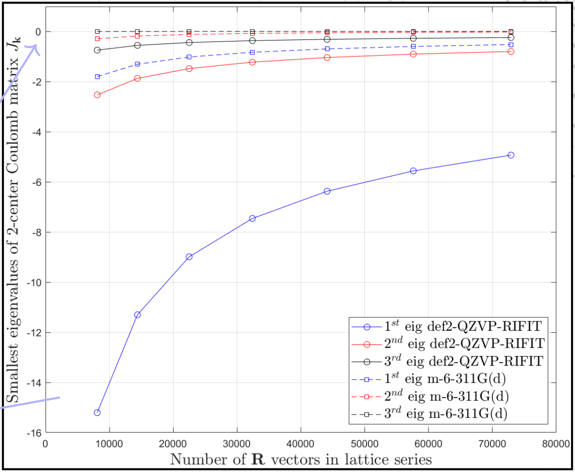


Fig.: hBN monolayer, $\boldsymbol{k} = (\frac{1}{15}, \frac{1}{15})$, $\sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\boldsymbol{R}} = 0$

Ewald substitution: automatic convergence

For any neutral and periodic density $\rho^{\text{latt}}(\boldsymbol{r}) = \sum_{\boldsymbol{R}} \rho^{\boldsymbol{R}}(\boldsymbol{r})$, an **Ewald potential function** $A(\boldsymbol{r})$ can be substituted

$$\int \rho^{\text{latt}}(\boldsymbol{r}')V(\boldsymbol{r}-\boldsymbol{r}')d\boldsymbol{r}' = \int \rho^{\boldsymbol{0}}(\boldsymbol{r}')A(\boldsymbol{r}-\boldsymbol{r}')d\boldsymbol{r}'$$

Saunders et al., *Molec. Phys.*, 77, 629 (1992)

Charge neutrality within the supercell $\Rightarrow$ Ewald substitution enabled

$\Downarrow$

For $\boldsymbol{k} \neq \boldsymbol{0}$ we obtain a *finite* and *exact* series:

2-center Coulomb matrix elements after Ewald substitution

$$J_{P,P'\boldsymbol{k}} = \sum_{\boldsymbol{R}}^{\text{SUP}} e^{i\boldsymbol{k}\boldsymbol{R}} \int \phi_P^{\boldsymbol{0}}(\boldsymbol{r})A(\boldsymbol{r}-\boldsymbol{r}')\phi_{P'}^{\boldsymbol{R}}(\boldsymbol{r}')d\boldsymbol{r}d\boldsymbol{r}'$$

At worst $\mathcal{O}(N_P^2 N_{\boldsymbol{k}}^2)$

Ewald potential functions

- **3D**:

Ewald potential function

$$A(\boldsymbol{s}) = \sum_{\boldsymbol{R}} \frac{\mathrm{erfc}(\sqrt{\gamma}|\boldsymbol{s}-\boldsymbol{R}|)}{|\boldsymbol{s}-\boldsymbol{R}|} + \frac{4\pi}{V} \sum_{\boldsymbol{G}\neq\boldsymbol{0}} \frac{1}{|\boldsymbol{G}|^2} e^{-\frac{|\boldsymbol{G}|^2}{4\gamma}+i\boldsymbol{G}\boldsymbol{s}}$$

3D series

$\gamma > 0$: arbitrary range-separation parameter

- **2D**:

Parry potential function [D. Parry, *Surf. Sci.*, 49, 433 (1975)]

$$A(\boldsymbol{s}) = \sum_{\boldsymbol{R}} \frac{\mathrm{erfc}(\sqrt{\gamma}|\boldsymbol{s}-\boldsymbol{R}|)}{|\boldsymbol{s}-\boldsymbol{R}|} - \frac{2\pi}{\mathcal{A}} \left[s_z\,\mathrm{erf}\left(\sqrt{\gamma}s_z\right) + \frac{1}{\sqrt{\pi\gamma}} e^{-\gamma s_z^2} \right]$$

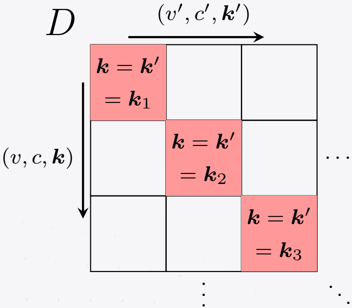
$$+ \frac{\pi}{\mathcal{A}} \sum_{\boldsymbol{G}\neq\boldsymbol{0}} \frac{e^{i\boldsymbol{G}\boldsymbol{s}}}{|\boldsymbol{G}|} \left[e^{|\boldsymbol{G}|s_z}\,\mathrm{erfc}\left(\frac{|\boldsymbol{G}|}{2\sqrt{\gamma}} + \sqrt{\gamma}s_z \right) + (s_z \leftrightarrow -s_z) \right]$$

2D series ✓

Divergences in the periodic GW-BSE

What happens with $J_{\boldsymbol{k}}$ at $\boldsymbol{k} = \mathbf{0}$?

- The commensurate supercell cannot be defined $\Rightarrow$ $J_{\mathbf{0}}$ **diverges**!
- This $\Gamma-$point case is realized for:
  1. The whole X matrix at $\boldsymbol{Q} = \mathbf{0}$
  2. The diagonal $\boldsymbol{k} = \boldsymbol{k}'$ blocks in D ($\forall \boldsymbol{Q}$)



Our solution:

1. Solve for $X(\boldsymbol{Q} \to \mathbf{0})$ $\Rightarrow$ non-analytical limit requires angular averaging
2. Unavoidable for D, but it's an artifact of the discretization of $\boldsymbol{k}-$space $\Rightarrow$ average over the $(2 \cdot \mathrm{dim})-$dimensional neighborhood around $\boldsymbol{k} = \boldsymbol{k}'$

Application to paradigmatic solids

FCC Argon

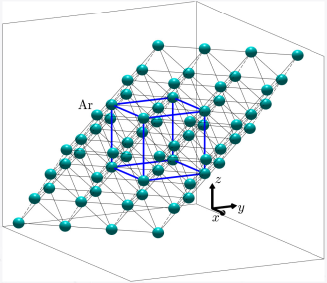


Fig.: FCC Argon lattice

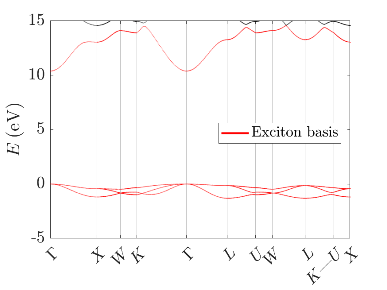
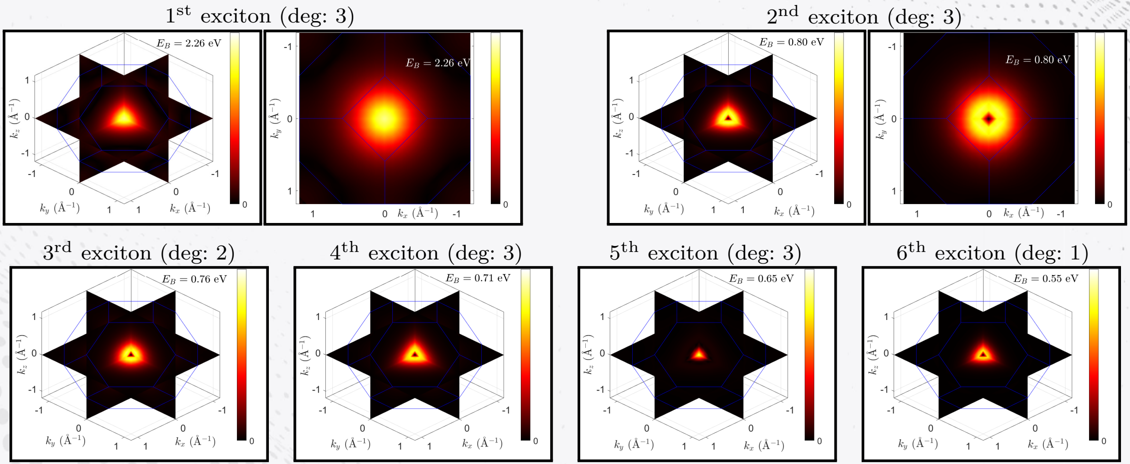


Fig.: Basis: ~def2-TZVPPD. Functional: HSE06

BSE convergence parameters

- $n_v = 3$, $n_c = 1$
- $\boldsymbol{k}-$grid: $22 \times 22 \times 22$ for BSE basis, $6 \times 6 \times 6$ for polarizab.
- Scissor correction: 3.86 eV (27 %)
- Metric: overlap
- Aux basis: def2-TZVPPD-RIFIT (enlarged)

FCC Argon - Exciton amplitude in $\boldsymbol{k}-$space



FCC Argon - Absorption

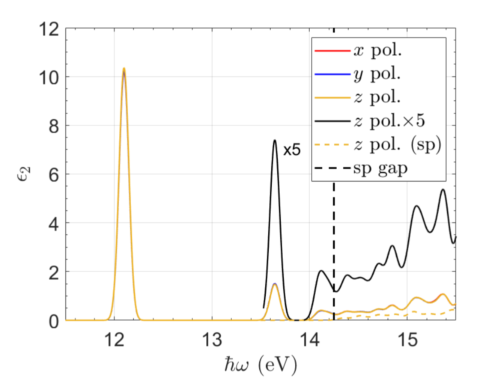


Fig.: **Present work**

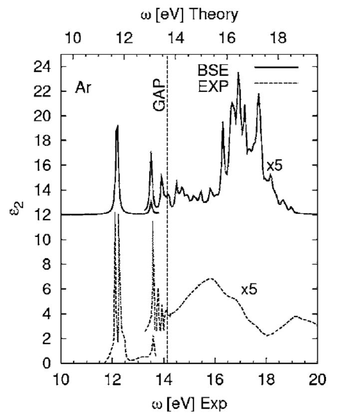


Fig.: From [1]

[1] S. Galamić-Mulaomerović, C.H. Patterson *Phys. Rev. B*, 72, 035127 (2005)

Boron Nitride (single-layer)

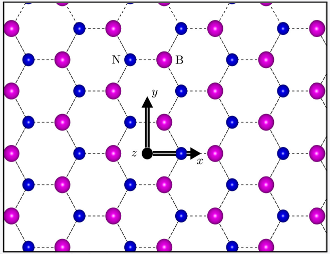


Fig.: 1L of hBN

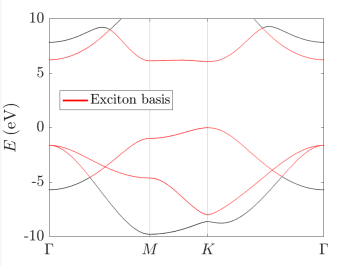


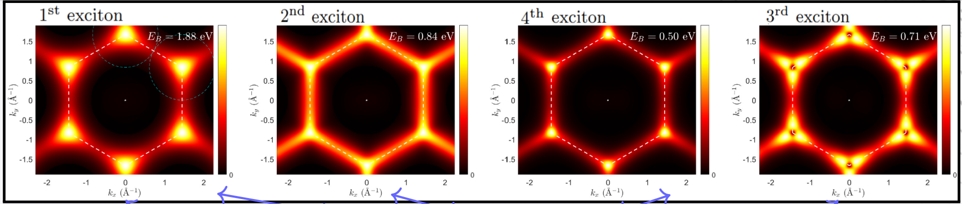
Fig.: Basis: ∼def2-TZVP. Functional: HSE06

BSE convergence parameters

- $n_v = 2$, $n_c = 1$
- $\boldsymbol{k}$−grid: 102 × 102 for BSE basis, 9 × 9 for polarizab.
- Scissor correction: 1.95 eV (24 %)
- Metric: overlap
- Aux basis: def2-QZVP-RIFIT

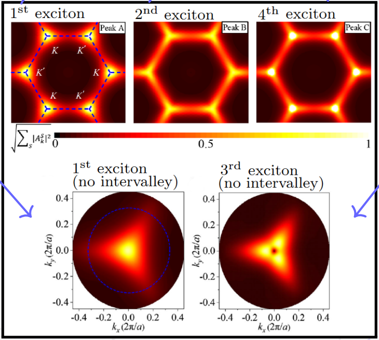
Boron Nitride - Exciton amplitude in $\boldsymbol{k}$−space

Present work



[1] Zhang, F. et al., *Phys. Rev. Lett.*, 128, 047402 (2022) [BerkeleyGW]

Zoom-in at a valley



Zoom-in at a valley

Boron Nitride - Absorption

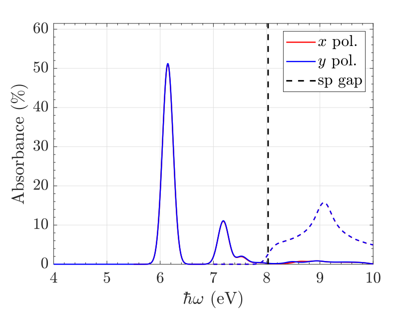


Fig.: **Present work**

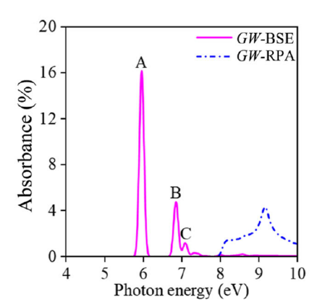


Fig.: From [1]

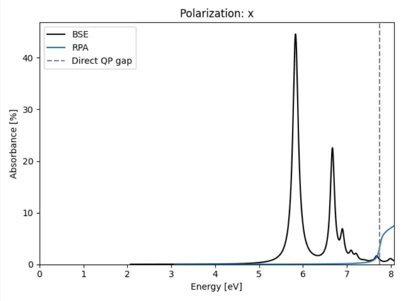


Fig.: From [2]

[1] Zhang, F. et al., *Phys. Rev. Lett.*, 128, 047402 (2022) [`BerkeleyGW`]

[2] Computational 2D Materials Database (C2DB) [`GPAW`]

Comparative of metrics for binding energies

Black Phosphorus (1L)

ω (a_0^{-1})	∞ (overlap)	∞	0.5	0.25	0.25
	Triplet	Singlet	Singlet	Singlet	Triplet
E_1	1.395	1.486	1.484	1.480	1.390
E_2	1.763	1.764	1.762	1.759	1.758
E_3	1.872	1.888	1.886	1.883	1.867
E_4	1.933	1.936	1.935	1.932	1.929
E_5	1.974	1.975	1.973	1.970	1.969
E_6	2.021	2.030	2.029	2.026	2.017
E_7	2.038	2.038	2.037	2.034	2.034

FCC Argon

ω (a_0^{-1})	∞ (overlap)	∞	0.5	0.5
	Triplet	Singlet	Singlet	Triplet
E_1 ($d = 3$)	11.821	12.100	12.087	11.808
E_2 ($d = 3$)	13.485	13.485	13.438	13.478
E_3 ($d = 2$)	13.511	13.527	13.480	13.504
E_4 ($d = 3$)	13.563	13.579	13.532	13.556
E_5 ($d = 3$)	13.620	13.644	13.598	13.614
E_6 ($d = 1$)	13.634	13.737	13.690	13.627

Differences of $\sim$meV between overlap and att. Coulomb metrics

$\Downarrow$

... but rather large auxiliary bases were used

Conclusions

M.A. García-Blázquez and J.J. Palacios, *Phys. Rev. Res.* 7, 013156 (2025)

- **Electronic correlations** are **essential** to reproduce optical measurements, e.g. absorption
  - GW-BSE more expensive but more accurate than TDDFT (@LDA, GGA)
  - GW-BSE could be comparable to hybrid TDDFT → extensive testing in solids is needed

- **Localized bases** are not "standard" but **suitable** for ab-initio description of excitons in solids

- **RI approximation** allows to significantly **reduce the scaling** and memory consumption
  - Choice of auxiliary basis is crucial
  - Overlap metric may be the optimal choice in some cases

- **Ewald substitution** yields exponential **convergence** of Coulomb lattice series
  - 2D systems are treated w/o replication with Parry potential
  - Must be employed in a supercell for electron-only problems (e.g. TDDFT, not DFT)